# Tactile World Models, Texture Simulation & Whole-Body Somatosensation

A comprehensive synthesis of our discussion: what a robot should feel, how to simulate it, what researchers have built, and a feasible route toward physically grounded touch for Physical AI.

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Includes: material representations and contact physics; 2026 tactile world models; visual-tactile simulation; differentiable mechanics; whole-body artificial skin; datasets and tactile foundation models; research hypotheses; phased experiments; and an annotated reading catalog.

## CENTRAL QUESTION

**Can an embodied AI system learn a reusable physical description of surface texture and contact, simulate what different bodies and sensors would feel, and use those predictions to act reliably in the real world?**

Evidence convention: Published work and reported experimental results are distinguished from design proposals and hypotheses. Example material parameter values are illustrative, not calibrated measurements. Source links appear in the numbered literature catalog.

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## How to use this report

Sections 1-4 define the problem and representation; 5-9 review simulators, learned models, sensing and data; 10-13 turn the literature into a testable research agenda; 14 gives the full literature map and reading order. The bibliography supplies direct clickable links.

# 1. Executive synthesis

Short answer: Tactile world models already exist. They predict future tactile or visuotactile observations from recent sensor readings, actions and sometimes force/torque signals. Examples are SPOTS, TacForeSight, TACO, FeelWorld and ContactWorld [39] [40] [41] [42] [43]. Their published demonstrations are mostly task-specific or sensor-specific; none establishes a universal, physically calibrated, whole-body tactile world.

A conventional robot simulator already models rigid-body geometry, inertia, contact, friction and sometimes deformable mechanics. But a single friction coefficient or soft-contact parameter does not reproduce the fine spatial and temporal signals of fabric, wood grain, skin or rubber sliding over a fingertip. Optical tactile simulators can generate sophisticated sensor images; electronic skins sense distributed contact; tactile encoders learn compact representations. These advances solve different parts of the problem.

Proposed long-term objective: Build a physically grounded somatosensory world model in which object materials and body sensors are modeled separately, a contact engine outputs pressure/shear/deformation/slip/vibration/heat, and a learned predictor improves dynamics forecasts and action selection. Whole-body touch is a long-term integration goal, not a prerequisite for a credible first experiment.

| What exists now | What is still open |
|---|---|
| Physics engines with friction, compliant contact and deformables | Transferable fine-texture, microslip and vibration across arbitrary surfaces |
| Tactile rendering for GelSight, DIGIT and similar sensors | Sensor-independent physical contact fields and validated tactile asset formats |
| Action-conditioned learned tactile / visuotactile prediction | Generalization to unfamiliar loads, motions, materials, sensors and embodiments |
| Partial or extensive robot-body electronic skin | An integrated, high-fidelity, human-scale simulated somatosensory body |
| Multisensory object datasets and tactile feature encoders | Jointly identifiable physical representations linked to repeatable measurement standards |

Strongest near-term hypothesis: With limited real-world measurements, a physically structured material representation plus a learned residual can predict held-out tactile interactions more robustly than a purely data-driven sensor-output predictor. This is a research hypothesis to test, not an established result.

# 2. Terminology and scope: physics engine vs. world model

| Term | Meaning | Role in this project |
|---|---|---|
| Contact-rich physics simulator | Computes contact forces, motion and deformation from bodies/materials/actions. | Grounded physical dynamics. |
| Tactile / haptic simulator | Maps contact events to tactile measurements or perceptual signals. | Synthetic sensations for robot learning. |
| Tactile world model | Learned action-conditioned forecast of tactile state (sometimes jointly with vision). | Anticipation, planning and control. |
| Somatosensory world model | Proposed broad phrase for distributed body touch plus physical prediction. | Long-term system framing, not a universal established definition. |
| Tactile material library | Reusable spatially and dynamically calibrated material descriptions. | Portable simulation assets. |
| Haptic foundation model | General pretrained model covering multiple touch/haptic settings. | Representation and transfer research direction. |

The safest concise project name is physically grounded tactile world model for the predictive component, with a tactile-aware physics and sensor simulation layer underneath it. The eventual full-body system can reasonably be described as a somatosensory world model.

## Why vision alone is not enough

Two surfaces can look identical yet differ in friction, stiffness, viscoelasticity, temperature and wear. Touch supplies latent information that determines whether a grasp slips, a surface scratches, an insertion jams or an object deforms. SPOTS specifically studies settings with visually similar objects but different physical properties and finds tactile signals are most valuable when the physical differences are visually ambiguous [39].

## Why “a 3D model plus friction” is not enough

Traditional simulation does include forces and friction; it is inaccurate to say world models or physics engines have only 3D meshes. The limitation is that many robotics pipelines do not render sensor-realistic, high-frequency, spatially distributed tactile experiences. A friction coefficient can help predict gross sliding; it cannot by itself recreate the woven microtexture of denim or the vibration spectrum of brushed metal.

# 3. What to represent in a touchable physical world

A touchable object needs more than a texture image. A useful physical description has geometry at several scales; local mechanical, tribological and thermal properties; and uncertainty about those properties. The robot also has its own compliant skin and sensor dynamics. These are distinct entities.

| Layer | State / parameters | Why it matters |
|---|---|---|
| Macro geometry | Mesh, signed distance field (SDF), collision geometry | Where and how the bodies meet. |
| Microgeometry | Height field, weave, pores, scratches, orientation maps | Microcontacts, roughness and dynamic excitation. |

| Layer | State / parameters | Why it matters |
|---|---|---|
| Friction / tribology | Static and kinetic friction, anisotropy, rate/load dependence | Sliding, stick-slip, incipient slip. |
| Compliance | Stiffness or Young's modulus, Poisson ratio, local anisotropy | Contact patch and pressure under load. |
| Damping / viscoelasticity | Rate-dependent losses and relaxation | Contact hysteresis and transients. |
| Adhesion | Attraction, tack and separation force | Soft and sticky surfaces. |
| Thermal | Conductivity, heat capacity and thermal contact | Why two equally warm materials feel different. |
| Vibration / acoustics | Frequency-dependent mechanical response | Fine texture during rubbing, tapping or impact. |
| Contact state | Normal pressure, shear traction, slip, deformation, temperature | What actually drives a simulated sensor. |

A basic material record might contain a mean friction coefficient and bulk stiffness. That describes a material category, not a complete tactile texture. For a surface parameterized by local coordinates (u,v), propose a spatial material field:

$$\mathbf{M(u,v) = [\ h(u,v),\ \mu_s(u,v),\ \mu_d(u,v),\ E(u,v),\ \nu(u,v),\ \eta(u,v),\ A(u,v),\ k_{th}(u,v),\ \dots\ ]}$$

Here h is microgeometry height; μ are friction parameters; E and ν describe elasticity; η is damping/viscoelastic behavior; A is adhesion; and $k_{th}$ is a thermal parameter. A richer implementation allows orientation, strain, speed, normal load, time, wear and environmental conditions to affect those values.

## Material is not sensation

The same object produces different signals when pressed, slid or tapped. Therefore, the sensation is a function of material + surface geometry + action + contact physics + the sensing body:

$$\mathbf{T_{sensor}(t) = Render_{sensor}(\ Contact(\ M,\ G,\ Body,\ a(0{:}t)\ )\ )}$$

Example tactile outputs range from an optical GelSight RGB/deformation image to a sampled pressure/taxel array or even a three-channel 128 × 128 normal/shear-force grid (the latter is an illustrative output design, not a universal sensor standard). A smooth glass surface is nearly flat, comparatively stiff and often frictionally uniform at the resolution of a coarse model. Denim has oriented weave microgeometry and direction-dependent interactions. Rubber may have strong compliance, adhesion and rate-dependent hysteresis. These are qualitative examples; numerical values must come from measurement.

## Illustrative tactile material asset

**Denim_12oz_014.tactile** (hypothetical file schema): spatial height/normal maps; weave period and direction; local friction parameters with uncertainty; normal/shear compliance; mechanical damping; temperature-dependent properties; calibration provenance; supported scales, ranges of loading/speed, and validation errors. Specific values such as a 0.4 mm period or friction 0.52, mentioned in our earlier discussion, were illustrative only, not empirical estimates for denim.

Other example material assets discussed: oak wood, brushed aluminum, corrugated cardboard, cotton, silicone rubber, smooth glass and human fingertip/forearm skin. Each would require standardized characterization before becoming an interchangeable simulation asset.

# 4. How texture becomes a tactile signal

The critical physical chain is: surface topography and material properties → contact mechanics → pressure/shear/slip/vibration/heat → sensor output. Fine tactile texture is typically dynamic: sliding excites frictional forces and vibrations, so surface roughness alone does not specify the experience.

## Normal and tangential contact

A point-contact approximation may report only an aggregate normal force. A touch-aware simulator should ideally produce a distributed contact patch:

$$C(u,v,t) = [\ P_n,\ \tau_x,\ \tau_y,\ \delta,\ v_{slip},\ T,\ q,\ \ldots\ ](u,v,t)$$

$P_n$ is pressure; $\tau$ are tangential tractions; $\delta$ is deformation; $v_{slip}$ local sliding velocity; $T$ is temperature; $q$ heat flux or related signal. The sensor renderer can then integrate or filter $C$ into pixels, taxels, magnetic readings or strain gauge signals.

## Slip is more than a binary label

Contact may transition from sticking to partial slip and then gross sliding. Incipient slip can begin at the edge of a contact patch before the object visibly moves. Realistic simulation may need friction that depends on speed, load, surface condition and direction. For texture, compare temporal force fluctuations and vibration spectra across scanning speeds, not just scalar friction coefficients.

## Human texture perception and vibration

Human-inspired research highlights the joint role of slowly varying deformation and fast vibration. One 2023 sensory-system paper measures both spatial pressure and high-frequency responses and demonstrates texture recognition [20]. Related material-identification work studies thermal and vibration cues, cross-modal texture reasoning and triboelectric effects [21] [22] [23] [24] [25]. These are useful for selecting which physical quantities to measure, but recognition accuracy alone does not establish that a simulator can synthesize touch.

## The “PBR for touch” analogy

Visual physically based rendering (PBR) commonly uses maps such as albedo, normal and roughness. The proposed analogy is a standardized physically based tactile material encoding microgeometry, friction, compliance, damping, adhesion, thermal behavior and uncertainty. Unlike ordinary visual textures, tactile assets must be specified with units, measurement protocols and validity ranges. The target is not a new file extension alone; it is a material representation whose predictions survive changes in action and sensor.

# 5. Physics engines and tactile simulation techniques

Different simulators solve different levels of fidelity. A high-quality robotics system will likely be multiscale: fast approximate dynamics for planning, selectively high-fidelity soft-contact simulation for calibration, and efficient learned residuals for texture or sensor-specific phenomena.

| Approach | Typical output | Strength / weakness |
|---|---|---|
| Rigid-body engine (MuJoCo, PhysX) | Object poses, contact forces, joints | Fast robot control; limited fine tactile detail without extensions. |
| Hydroelastic contact (Drake and related) | Contact surfaces, distributed normal pressure | Closer to skin pressure fields; details still depend on material/discretization. |

| Approach | Typical output | Strength / weakness |
|---|---|---|
| Sensor image rendering (TACTO, Taxim) | GelSight/DIGIT-like tactile images or markers | Realistic for specified optical sensors; not automatically a general material model. |
| GPU visuotactile systems (TacSL) | Tactile images/contact-force maps at scale | Useful for learning; modeling assumptions and calibration remain crucial. |
| FEM / SOFA | Spatial deformations, stresses, strain | Rich soft-body mechanics; computationally expensive. |
| MPM / differentiable simulation | Deforming material state and gradients | Powerful for inverse material identification; harder to validate across texture scales. |

## Optical tactile rendering: TACTO, GelSight in Gazebo, Taxim, Tactile Gym 2.0

TACTO demonstrates fast configurable rendering for high-resolution optical tactile sensing [4]. GelSight-in-Gazebo work introduces synthetic GelSight-style contact images in conventional robotics simulation [5]. Taxim uses an example-based model, including optical response and elastomer/marker displacement, to better reproduce GelSight observations [6]. Tactile Gym 2.0 extends optical tactile learning and sim-to-real evaluation across tactile sensor designs [7].

These systems are crucial, but optical realism and physical validity are separate questions. A generated tactile RGB image may look plausible while encoding inaccurate friction, microscopic shear or vibration. Sim-to-real validation should compare measured physical outcomes, not only image similarity.

## TacSL and newer GPU pipelines

TacSL integrates accelerated visuotactile simulation and robot-learning workflows in a GPU ecosystem [8]. Isaac Lab’s experimental visuotactile module can calculate normal/shear force fields with an elastomer response [49]. These systems may make large-scale policy training practical, but the user’s proposed ambition still requires broader material parameterization, better experimental calibration and whole-body coverage.

## HydroelasticTouch and Drake

Hydroelastic formulations replace one isolated contact point with a surface carrying pressure information. HydroelasticTouch applies this principle to tactile rendering [9], while Drake documents compliant/hydroelastic contact mechanics [10]. NVIDIA’s emerging hydroelastic contact tools pursue similar distributed contact primitives [50]. Spatial contact matters because fingertip and skin observations depend on where forces are applied, not only their total magnitude.

## High-fidelity elastomer mechanics

FEM-based approaches can model the elastic skin geometry, marker movement and internal stress much more explicitly, though at a significant computational cost. The 2026 marker-level 3D deformation study exemplifies physics-rich visuotactile sensor modeling [11]. This makes it valuable as a reference simulator to calibrate faster tactile approximations.

### Simulation backends in the discussion

MuJoCo is a strong starting point for control and contact-rich robotics [12]. Drake offers well-developed compliant-contact machinery. Isaac/PhysX provides scalable GPU robot-learning infrastructure [48]. SOFA is useful for detailed deformable bodies and virtual prototypes. Taichi/MPM provides differentiable continuum-mechanics experiments. A plausible design uses an existing physics engine as the dynamics backbone and adds a portable tactile contact/material layer, instead of rewriting rigid-body simulation from scratch.

## 6. Inverse physics: learning material properties from touch

The core data-to-world question is whether you can infer physical material parameters from known actions and measured contact responses. This is a form of inverse mechanics / system identification, distinct from simply learning to classify textures.

$$\theta^* = \arg\min_{\theta} \Sigma_{a \in A} \, d( \, Sim(M(\theta), a), \, T_{real}(a) \, ) + \lambda \, R(\theta)$$

Here θ contains physically meaningful model parameters; A is a set of probing actions; d compares predicted and observed signals; and R regularizes ambiguous solutions. A differentiable simulator can provide gradients, but even nondifferentiable systems can be fit through search or surrogate models.

### Relevant differentiable physics work

ChainQueen applies differentiable material-point simulation to soft robotics [13]. DiffTaichi develops differentiable programming for physical simulation [14]. PlasticineLab uses differentiable mechanics for deformable-body manipulation [15]. Dojo investigates differentiable robot contact/friction and system identification [16]. IPC and related work improve robust treatment of deformable-body contact [17]. None alone solves tactile material identification, but they supply mathematical infrastructure for it.

### Identifiability is a major open issue

The same force trace may be reproduced by different combinations of sensor compliance, object compliance, friction and geometric microstructure. Accurate labels therefore need controlled normal loads, independent force/torque measurement, known motion, calibration specimens, repeat measurements, and multiple action types. A model that fits seen pressing trajectories may still fail under new sliding speeds. Sensor aging, fabrication variability, temperature and wear must be separated from material properties.

### Hybrid simulation plus learned residual

An effective architecture may use: (i) a physical material/contact model for causally meaningful pressure and shear; (ii) an observation model for a particular sensor; and (iii) a learned residual to capture microtexture, optical effects, unmodeled vibration or solver bias. In abstract form:

$$T_{pred} = Render( \, Contact(\theta, a) \, ) + Residual_{ML}(history, a)$$

The residual should remain limited and tested out of distribution, otherwise the model risks reverting to a black-box texture predictor. An uncertainty estimate should flag conditions beyond the calibrated loading, speed, sensor or environmental range.

# 7. Multisensory object representations: ObjectFolder and related work

ObjectFolder is among the closest precursors to the idea of a downloadable touchable object. Its implicit representations let virtual objects produce visual, auditory and tactile observations; ObjectFolder 2.0 scales the concept to 1,000 common objects and studies sim-to-real transfer [18] [19].

The crucial architectural difference in the proposed project is to distinguish a physical object field from a sensor-dependent tactile response. An object-specific tactile network can generate the appropriate GelSight image without necessarily determining a stable set of coefficients governing friction, wear, compliance or force under arbitrary new motions. A more transferable system would separate:

Geometry + intrinsic material field → contact mechanics → canonical physical contact state → sensor renderer → observed tactile stream.

This structure could allow a new tactile sensor, body skin or robot to interact with the same material asset. It also enables benchmarks for each link in the chain rather than evaluating only the final image. Vision-touch cross-modal prediction [26] provides a useful complementary learned baseline.

# 8. Existing tactile world models: what they do and how

The five models discussed in our October conversation establish that learned tactile prediction already exists. They are not equivalent to a complete virtual physical world. Most learn regularities from recorded interaction trajectories rather than reconstructing the full underlying microscale material mechanics.

## The generic training and usage loop

- Collect synchronized demonstrations of pushing, grasping, sliding, insertion or wiping with cameras, tactile sensors, motor actions, robot state and often wrist force/torque.
- Encode tactile images or taxels into a lower-dimensional latent representation; encode video, proprioception and candidate actions.
- Predict future tactile signals, latent states, visual frames, contact events or slip across a limited horizon.
- Act by planning against these futures, detecting impending instability, or synthesizing recovery examples for policy learning.

$$p\left( z_{t+1:t+H}^{\text{touch}}, z_{t+1:t+H}^{\text{vision}} \mid \text{history}_t, a_{t:t+H} \right)$$

For intuition, consider a robot holding a glass: it senses the current grasp, imagines loosening its fingers, forecasts falling contact pressure and possible slip, and tightens its grip. This is predictive sensor modeling even if the system never explicitly estimates the glass friction coefficient.

## SPOTS — predicting touch and vision together

What: SPOTS jointly predicts future visual and tactile frames from robot-pushing trajectories. It uses modality-specific pathways with cross-modal interaction. Why: touch adds information especially when objects look the same but differ physically. Boundary: the experiments involve pushing and particular tactile measurements, not arbitrary materials or whole-body touch [39].

## TacForeSight — anticipating local contact dynamics

What: TacForceWM encodes dual-finger tactile observations into compact latent states and predicts their near-term evolution conditioned on wrist force/torque; the resulting prediction informs a tactile-conditioned policy. Tasks: wiping, swiping, adjustments and insertions, including disturbances. Boundary: short-horizon tactile latent forecasting, not a general reusable friction/texture asset [40].

## TACO — imagined visuotactile corrections

What: a Recognize–Imagine–Label process identifies failure-adjacent robot states, generates possible corrective video/tactile trajectories, labels candidate actions and filters for plausible recovery. Reported metric: average task score improved from 0.375 to 0.825 after two post-training iterations in its evaluated setting. Boundary: the metric is paper-specific and does not establish general material simulation [41].

## FeelWorld — contact, force features and slip

What: jointly predicts visual latents and three tactile components: contact occurrence, a compact force-related 3D tactile latent, and slip state. Contact-gated attention emphasizes touch when contact actually occurs, and predicted states support planning. Reported metric: 81.7% mean zero-shot planning success in the authors' evaluated tasks. Boundary: task- and representation-specific empirical result, not a transferable whole-body material library [42].

## ContactWorld — which representations actually help?

What: evaluates different visual/tactile representations across 12 contact-rich tasks; emphasizes spatial structure and temporal continuity. Reported metrics: average planning success of 32.1% with point-cloud observations versus 36.1% when tactile force-field information is combined, with vision-only camera baselines lower. Boundary: controlled benchmark results; the importance of structure does not prove a universal contact simulator [43].

| Project | Prediction / method | Takeaway for your direction |
|---|---|---|
| SPOTS | Joint tactile + image future in pushing | Touch matters most when physics is visually ambiguous. |
| TacForeSight | Wrist-force-conditioned short-horizon tactile latent | Contact forecasts can guide high-frequency control. |
| TACO | Generate, filter and label recovery trajectories | World models can create contact-aware policy training data. |
| FeelWorld | Contact, compact force features, slip + visual latent | Separate physical contact events from image appearance. |
| ContactWorld | Compare visual/tactile latent representations for planning | Spatial structure and cross-modal alignment matter. |

Comparison warning: Reported planning successes, task scores and prediction errors are not directly comparable across different papers, sensors or tasks. Read the experimental protocols before treating any number as a ranking.

# 9. Tactile foundation models and cross-sensor representations

A world model should not necessarily process millions of raw taxel or image values at each step. Tactile representation learning compresses sensor signals into a meaningful state z_touch that can be fused with vision, body pose, robot actions, and force/torque.

**Raw tactile stream → Tactile encoder → $z_{touch}$ → Action-conditioned dynamics model**

## Sparsh

Sparsh learns broad vision-based tactile features from unlabeled touch images and evaluates these representations on tactile tasks. Its importance is methodological: tactile perception can be pretrained rather than learned from scratch for each grasp or surface. However, good downstream scores do not imply the learned embedding corresponds one-to-one with intrinsic physical coefficients [27].

## AnyTouch and AnyTouch 2

AnyTouch aligns tactile representations across heterogeneous visuotactile sensors, suggesting that a common latent space could allow transfer across hardware. AnyTouch 2 extends the program toward temporal/contact dynamics and force-aware information [28] [29]. A sensor-agnostic latent is valuable, but it is not automatically a sensor-independent physical contact field. The proposed architecture should keep those concepts separate.

## Connecting Touch and Vision

Cross-modal prediction research shows visual cues can predict tactile-related information and tactile cues can inform visual inference [26]. This supports a multisensory world model with a shared physical latent state, but action-conditioned validation remains essential: recognizing that a cloth is rough is not the same as predicting how force and vibration change during a new stroke.

## The gap between recognition and prediction

| Learning objective | Example target | Scientific limitation |
|---|---|---|
| Texture recognition | Classify denim vs cotton | Labels do not describe new motions or forces. |
| Tactile representation | Compress to transferable z_touch | Latents may be hard to interpret physically. |
| Tactile prediction | Predict future sensor readings | Could overfit a known sensor/task distribution. |
| Physical parameter estimation | Estimate friction/compliance from many probes | Parameters may be ambiguous without independent ground truth. |
| Generative physical tactile world | Predict contact + arbitrary sensor outputs | Requires joint mechanics, calibration, action diversity and validation. |

# 10. Human-like whole-body tactile sensing

Human somatosensation goes beyond fingertip contact. Skin conveys pressure, motion, vibration, stretch, warmth and cold; the body simultaneously integrates proprioception and motor predictions. A robot approximating human-scale touch needs distributed sensors, calibration, skin geometry, bandwidth management and a body-aware perception/control system.

## What a full-body state might contain

$$\text{Skin}(s,t) = [\ P,\ F_x,\ F_y,\ F_z,\ \text{shear},\ \text{slip},\ \text{vibration},\ \text{temperature},\ \text{deformation}\ ]$$

Here s indexes points on the robot's physical body surface. No existing affordable skin necessarily measures all components directly at high resolution. Some must be estimated from nearby measurements or from a mechanistic skin model. The target is a distributed sensing and state-estimation system, not simply a dense matrix of force values.

## Whole-body humanoid artificial skin

A self-organizing modular multimodal skin has been demonstrated on a humanoid body in earlier research [30]. Newer whole-body tactile-control work uses distributed pressure/force, proximity, vibration or temperature readings to inform compliant contact behavior [31]. These show that whole-body tactile sensing and control are active research areas, though they do not provide a universal physics-calibrated texture world model.

## Scalable electronic skin: RobotSweater

RobotSweater develops a customizable knitted pressure-sensitive skin that can cover curved and large areas and detect/localize contact [32]. Its relevance is scalability: a humanoid cannot afford an optical camera and elastomer assembly beneath every square centimeter. Lower-resolution large-area pressure skin may coexist with dense, high-bandwidth fingertips.

## ReSkin, AnySkin and D'Manus

ReSkin studies flexible magnetic tactile sensing with replaceable sensing surfaces [35]. AnySkin explores a practical plug-and-play direction and transfer across replacement skins [36]. D'Manus distributes magnetic sensing across the fingers and palm and offers relevant integrated hand/simulation ideas [37]. These works motivate a central standardization problem: signals from two skins may differ because of manufacturing, electronics or calibration even on the same object.

## A realistic hierarchical body design

- Fingertips: high-spatial-resolution normal/shear contact, microslip, vibration and local deformation.
- Palms / feet: broad contact patches, load distribution, shear direction and stability.
- Arms / torso / legs: coarser large-area pressure, proximity and protective contact sensing, with local higher-resolution zones where needed.
- Central representation: map every measurement to registered body geometry, time, calibration metadata and an uncertainty estimate.

This yields heterogeneous resolution, which is likely more practical than identical "human-like" sensing everywhere. It also introduces a modeling challenge: sensors across the body must have a shared coordinate system and comparable units even when their sensing physics differs.

### Research reviews on whole-body skin

The 2025 humanoid e-skin review summarizes receptor modalities and integration issues [33]. A separate review surveying many human-inspired e-skin approaches highlights fragmented specifications and standards [34]. Both support the need for common data schemas and calibration protocols; neither resolves the material simulation problem by itself.

## 11. Human tactile demonstrations and useful data

One route is to collect contact-rich human movement with an instrumented glove or suit rather than only industrial robot trajectories. The Feel the Force project uses force/tactile information from human demonstration to inform robot action [38]. The broader research idea is to observe: human hand/body motion, video, time-synchronized contact, normal/shear force, high-frequency vibration, and the identity and measured geometry of the touched object.

### What data should be collected?

| Data stream | Minimum logging expectation | Reason |
|---|---|---|
| Actions and motion | Pose, velocity, contact direction, trajectories, timebase | Makes tactile response action-conditioned. |
| Forces | Independent normal and tangential force, torque when possible | Anchors calibrated mechanics. |
| Tactile | Raw sensor channels and processed maps | Enables later model changes and auditing. |
| Visual / geometry | Synchronized video, body/object meshes or scans | Defines where and what is touched. |
| Material/sample metadata | Specimen ID, batch, fabrication, thickness, humidity, age | Separates variation from true physical properties. |
| Sensor metadata | Sensor ID, replacement, calibration, temperature, resolution | Tests transfer across hardware. |
| Repeat trials | Multiple loads, speeds, directions, days, operators | Tests reproducibility and uncertainty. |

The earlier discussion considered pressing, sliding, twisting, tapping, gripping, incipient slip and disturbance/recovery. For this specific material-modeling objective, structured loading and sliding protocols are more informative than unstructured hours of demonstrations. Collect both what happens and what controlled action caused it.

### Sensor and object diversity: do not confuse variation sources

To measure transfer, vary materials, object exemplars, sensor units and fabrication batches deliberately. More objects are not a substitute for multiple replacement sensors; multiple sensors are not a substitute for unseen sliding speeds or normal loads. ReSkin/AnySkin motivates explicit sensor-interchange experiments, and the discussion noted that learned transfer can still leave physical-scale force bias. Consequently any self-supervised alignment should be supplemented by independently measured force labels before it is interpreted as metrological calibration [35] [36].

### Data rights, provenance and standardized datasets

A commercially useful dataset needs explicit rights for collection and licensing, object/material provenance, sensor calibration records and clear measurement units. Preserve original time-stamped raw signals so future tactile encoders and contact models can be retrained. Synchronization should be evaluated quantitatively; a sensor supplier's advertised time alignment does not itself establish end-to-end measurement accuracy.

## 12. Integrated architecture: from materials to world models

The proposed design separates what belongs to the world, what belongs to the robot body, and what the learning system predicts. This is the major integration concept throughout our discussion.

| Component | Input | Output |
|---|---|---|
| A. Physical world asset | Mesh/SDF; material field; surface microgeometry; environment | Interacting physical surface. |
| B. Robot body / action | Body mesh; compliance; motion; command; sensor placement | Contact kinematics. |
| C. Contact engine | A + B | Pressure, shear, deformation, slip, vibration/heat estimates. |
| D. Sensor renderer | Canonical contact field + sensor calibration | GelSight image, taxels, magnetic signal, pressure map, etc. |
| E. Multimodal encoder | Rendered or real touch; vision; proprioception | Aligned latent state with uncertainty. |
| F. World model + policy | Latent state and candidate actions | Future contact predictions and selected action. |

### World/skin decoupling is essential

Keep the object's intrinsic properties separate from the sensor response. Do not train one object→GelSight image network and call that a sensor-independent material model. The most portable interface is a documented contact field with spatial coordinates, standard units, time resolution, numerical tolerances and calibration assumptions.

### Two fidelity modes

- Fast training mode: approximate contact patches, learned vibration residuals, GPU-parallel sensing, coarse body taxels. Optimized for large-scale policy rollouts.
- Accurate validation mode: higher-resolution microgeometry, nonlinear contact, FEM reference simulations, higher sampling rates and detailed sensor calibration. Optimized for measuring error and identifying parameters.

This split is a practical way to reconcile a world-model training loop with the physical complexity of fine texture. A full-FEM, whole-body millisecond-by-millisecond world may be computationally prohibitive; the model should spend precision where it affects action.

### A causal physical state vs. a latent state

A latent z_touch can be excellent for predicting slip while encoding friction and deformation in uninterpretable combinations. A physical state provides units and more explicit counterfactuals but may miss sensor artifacts or complex tribology. The strongest strategy may be to retain both: physical parameters for intervention and an expressive latent for residual dynamics. The models must be independently evaluated against held-out actions and samples.

## 13. Research gap and technical obstacles

The central research gap is not “no tactile world models exist.” It is the absence, among the systems discussed here, of a broadly validated, reusable stack that combines intrinsic tactile material assets, realistic dynamic contact, multiple sensor modalities, and high-coverage body sensing with action-conditioned planning.

| Obstacle | Why difficult | What would count as progress |
|---|---|---|
| Physical identifiability | Different parameter combinations yield similar contact traces. | Recover stable physical parameters from diverse probes. |
| Fine texture / bandwidth | Microtexture excites fast contact dynamics. | Correct vibration spectra across held-out speeds. |
| Sensor bias | Skin replacement and calibration shift force scale. | Transfer to new sensor units with limited recalibration. |
| Scale separation | Whole-body touch vs millimeter/micrometer contacts. | Selective multiresolution contact with bounded error. |
| Data diversity | Too few loads, speeds, materials or batches. | Proper held-out axes and documented coverage. |
| World model drift | Autoregressive predictions accumulate error. | Long-horizon stability and calibrated uncertainty. |
| Contact and slip | Partial slip, adhesion and nonlinear deformation. | Predict onset, contact patch shape and recovery. |
| Proof of robot utility | Good sensor-image metrics may not help policies. | Better grasp/push/insertion success with tactile model. |

### What to avoid claiming

- Do not claim existing physics engines model no friction; they generally do at several degrees of fidelity.
- Do not equate recognition embeddings, sensor image synthesis, contact mechanics and whole-body somatosensation.
- Do not claim a learned predictor “understands material” merely because it forecasts one familiar motion.
- Do not equate cross-sensor feature alignment with physical force calibration.
- Do not interpret one paper’s task score as directly comparable to another paper’s success rate.
- Do not claim no one has explored tactile world models: SPOTS, TacForeSight, TACO, FeelWorld and ContactWorld are direct counterexamples.

# 14. A feasible first research study: calibrated tactile material transfer

A strong first paper or startup prototype is not a full-body human-touch simulator. It is a controlled demonstration that a learned physical surface model predicts tactile behavior under new actions better than existing alternatives. The study needs reproducible measurements, baselines and a clear failure analysis.

## Research question and hypothesis

Question: Can a reusable, physically parameterized tactile material representation learned from a small set of controlled interactions accurately predict force, vibration, slip and tactile observations under previously unseen contact conditions?

Hypothesis: With equal training data, a hybrid physics+residual model will generalize more robustly across unseen normal forces, scan speeds, directions and sensor instances than a sensor-specific black-box predictor.

## Experimental setup

- Materials: 3-5 well-characterized surfaces with contrasting physics: smooth polymer/glass-like surface, silicone rubber, denim or woven fabric, corrugated cardboard and brushed metal. Use repeat specimens.
- Actuator: robot arm or controlled linear stage with a repeatable probe trajectory; log commanded and measured motion.
- Sensors: one tactile fingertip/probe plus independent force/torque measurement; if possible a high-sample-rate vibration sensor and visible markers for motion tracking.
- Trials: normal indentation and release, speed-controlled sliding, multiple normal loads, two or more scan directions; add taps/twists only after a reliable baseline.
- Synchronization: all streams on a checked time base with known frame/sample rates; archive raw calibration and contact geometry.

## Baselines and experimental comparisons

| Arm | Method | Expected behavior to evaluate |
|---|---|---|
| A: Purely learned | Action-conditioned network directly forecasts tactile/force signals. | Potential strong interpolation; test extrapolation and sensor replacement. |
| B: Physics only | Fit simple friction/compliance and deterministic contact. | Interpretability; may miss dynamic fine texture. |
| C: Hybrid (primary) | Fit physical parameters + learned residual. | Does physical structure improve transfer? |
| D: Strong literature baseline | Adopt/reproduce relevant representation or prediction method. | Avoid beating only deliberately weak baselines. |

## Held-out tests

- Unseen combinations of normal force and scan speed; avoid only random train/test trial splits.
- New scan directions and contact locations on the same material.
- Different manufactured specimens and different days, to test physical variability.
- Replacement tactile sensor or separately calibrated second unit, to test sensor transfer.

- If feasible, a new probe shape or an independent manipulation task for downstream validation.

### Metrics and failure analysis

Report calibrated normal/shear force error; contact-area or pressure-field error; slip-onset timing and detection; frequency-domain vibration similarity; temporal prediction robustness; sensor transfer/calibration cost; and real downstream grasp, push or insertion success. Include per-material and per-held-out-condition breakdowns, error bars across repeated trials, and ablations removing microgeometry, friction, physical supervision or learned residuals.

### Milestones, in sequence

- Phase 1 - Measurement: get one probe, one force reference and one material to produce repeatable press/slide traces; verify calibration.
- Phase 2 - Contact baseline: fit a simple friction/compliance model and predict an untouched trial; characterize the dominant errors.
- Phase 3 - Material comparison: expand to 3-5 surfaces and controlled ranges of speed, load and direction.
- Phase 4 - Learning baselines: compare black-box predictor, pure physics and hybrid residual under identical held-out tests.
- Phase 5 - Transfer: test a new specimen/sensor instance and demonstrate improvement in at least one robot task.
- Phase 6 - Reuse: export machine-readable material assets and connect them to a simulator independent of the original hardware.

Publication target: a falsifiable contribution to tactile representation and inverse physical modeling. A credible negative result that explains non-identifiability or cross-sensor failure can also be scientifically meaningful.

## 15. Long-term roadmap and strategic implications

### Three progressively harder deliverables

| Stage | Deliverable | Core proof |
|---|---|---|
| 1. Tactile material database | Reproducibly calibrated surfaces with provenance and uncertainty. | Predicts held-out tactile interactions. |
| 2. Somatosensory simulation layer | Canonical contact fields and adapters for several tactile devices / body regions. | Sensor- and embodiment-transfer in simulator and hardware. |
| 3. Tactile world model | Predictive vision+touch+action model supporting robot control. | Improves performance on unfamiliar contact-rich tasks. |

### What could be commercially defensible?

The central value proposition is not collecting enormous unstructured touch recordings. It is turning well-measured physical interactions into reusable, validated simulation-ready tactile material assets, with calibration procedures, provenance and known generalization bounds. A secondary offering could be standardized sensor data adapters and tactile simulation/robot-learning benchmarks. A third offering, later, could be a broad multimodal world model.

### Research vs. business claims

A material database is only useful to customers if synthetic tactile training demonstrably improves a real manipulation outcome (success rate, drops, excess force, tolerance to object variation or adaptation cost). This requires controlled tests against comparable vision-only, force-only and raw-tactile baselines. Market value, exclusivity, demand and willingness to pay were not established by the papers discussed; they need direct customer validation.

### Best positioning for PhD research

A research program should emphasize an identifiable, testable scientific question rather than only a product vision. A strong first submission would include a dataset with rigorous ground truth, a physically interpretable model or a new benchmark, controlled ablations and held-out transfer tests. Whole-body tactile intelligence then becomes a coherent long-term research arc connecting contact physics, machine perception and robot learning.

### Open questions raised by the conversation

- How many independent physical parameters can touch actually identify, and under what action diversity?
- Is microgeometry essential, or can a low-dimensional speed/load-dependent response model suffice for robot policy training?
- Which physical quantities must be modeled explicitly, and which can be learned as a residual?
- How does calibration error vary across tactile skins, fabrication batches, days and sensor types?
- What is the cheapest sensing protocol that achieves reliable cross-action transfer?
- Can a contact-field representation reproduce different sensors without per-object retraining?
- Does pressure/shear/vibration fidelity lead to measurable success on unfamiliar objects, or are simpler tactile cues sufficient?
- How should a humanoid allocate sensing density across fingertips, palms, feet and torso?

## 16. Research reading sequence and literature map

The following sequence preserves the reading plan discussed earlier, but organizes it as prerequisites rather than ranking every paper by importance. The annotated literature catalog after this section includes all significant items referenced in the conversation.

### Priority sequence: 25 works

Understand what touch sensors measure. GelSight [1]; Improved GelSight [2]; ReSkin [35]; AnySkin [36]

Extend to larger tactile embodiment. Whole-body humanoid skin [30]; RobotSweater [32]

Learn to synthesize tactile signals. TACTO [4]; Taxim [6]; TacSL [8]; HydroelasticTouch [9]; Drake contact [10]

Learn contact and inverse physics. ChainQueen [13]; DiffTaichi [14]; Dojo [16]; IPC [17]

Understand multisensory objects. ObjectFolder [18]; ObjectFolder 2.0 [19]; Connecting Touch and Vision [26]

Learn tactile representations. Sparsh [27]; AnyTouch [28]; AnyTouch 2 [29]

Study modern world models. SPOTS [39]; TacForeSight [40]; TACO [41]; FeelWorld [42]; ContactWorld [43]

This extends the original 25-paper progression to include FeelWorld and ContactWorld, which were subsequently discussed in the October follow-up. The original sequence also included D’Manus and an emerging haptic foundation-model perspective as adjacent follow-up reading.

### High-value review papers

Start from the historical reviews Tactile Sensing for Dexterous In-Hand Manipulation [44] and Tactile Sensing in Dexterous Robot Hands [45], then the 2026 survey Vision-Based Tactile Intelligence for Robotics [46] and the 2026 perspective on Haptic Foundation Models [47]. Add the 2025 humanoid e-skin reviews [33] [34] for whole-body sensing.

## 17. Annotated bibliography and direct links

This catalog captures the research papers, project pages, key simulators and surveys mentioned across our discussion, grouped by topic. Links lead to primary papers, publishers, authors' project pages or technical documentation where available. Project pages and documentation are clearly distinguished from peer-reviewed papers. Preprints and publication status can change; always check the linked source for its latest version.

### Sensors / GelSight

[1] GelSight: High-Resolution Robot Tactile Sensors for Estimating Geometry and Force (2017) — Foundational high-resolution optical sensing of local geometry and force. Open source ↗

[2] Improved GelSight Tactile Sensor for Measuring Geometry and Slip (2017) — Gel deformation and slip sensing in a vision-based tactile interface. Open source ↗

### Manipulation / tactile learning

[3] The Feeling of Success: Does Touch Sensing Help Predict Grasp Outcomes? (2017) — Tactile sensing for predicting grasp outcomes. Open source ↗

### Tactile simulation

[4] TACTO: A Fast, Flexible, and Open-Source Simulator for High-Resolution Vision-Based Tactile Sensors (2020/2021) — Sensor-oriented synthetic visuotactile observations. Open source ↗

[5] Generation of GelSight Tactile Images for Sim2Real Learning (2021) — GelSight-style image generation in simulation. Open source ↗

[6] Taxim: An Example-Based Simulation Model for GelSight Tactile Sensors (2022) — Data-calibrated optical rendering, elastomer deformation and marker motion. Open source ↗

[7] Tactile Gym 2.0 (2022) — Sim-to-real robot learning with multiple optical tactile sensor designs. Open source ↗

[8] TacSL: A Library for Visuotactile Sensor Simulation and Learning (2025) — Accelerated visuotactile simulation and learning infrastructure. Open source ↗

### Contact mechanics / tactile simulation

[9] HydroelasticTouch (2025) — Hydroelastic pressure-distribution-based tactile modeling. Open source ↗

### Physics engines / contact

[10] Drake: Compliant / Hydroelastic Contact Modeling Documentation (ongoing) — Distributed contact surface rather than only isolated point contact. Open source ↗

### Soft-contact / FEM

[11] High-Fidelity Marker-Level 3D Deformation Simulation of Visuotactile Sensors (2026) — Physics-rich elastomer and marker displacement simulation. Open source ↗

### Physics engines

[12] MuJoCo: A Physics Engine for Model-Based Control (2012) — Fundamental rigid-body and contact simulator for control. Open source ↗

## Differentiable physics

[13] ChainQueen: A Real-Time Differentiable Physical Simulator for Soft Robotics (2019) — Differentiable material-point simulation for soft-body modeling and optimization. Open source ↗

[14] DiffTaichi: Differentiable Programming for Physical Simulation (2020) — Framework for differentiating through physics simulation. Open source ↗

[15] PlasticineLab: A Soft-Body Manipulation Benchmark with Differentiable Physics (2021) — Soft body manipulation and differentiable physics benchmark. Open source ↗

[16] Dojo: A Differentiable Physics Engine for Robotics (2022) — Contact/friction-aware differentiable robot dynamics and system identification. Open source ↗

## Contact mechanics

[17] Incremental Potential Contact (IPC) (2020) — Stable simulation of contact for deformable geometry. Open source ↗

## Multisensory objects

[18] ObjectFolder: A Dataset of Objects with Implicit Visual, Auditory, and Tactile Representations (2021/2022) — Implicit multisensory object assets for tactile, audio and vision queries. Open source ↗

[19] ObjectFolder 2.0: A Multisensory Object Dataset for Sim2Real Transfer (2022) — 1,000 virtual objects with visual, auditory and tactile representations. Open source ↗

## Texture perception

[20] A Robotic Sensory System with High Spatiotemporal Resolution for Texture Recognition (2023) — Combines spatial pressure and dynamic high-frequency texture cues. Open source ↗

## Texture and materials

[21] Material Recognition Using Tactile Sensing (2018) — Tactile vibration and thermal cues for classifying materials. Open source ↗

[22] Multimodal Material Identification through Recursive Tactile Sensing (2018) — Material identification using multiple tactile sensing modalities. Open source ↗

[23] Human-Touch-Inspired Material Recognition for Robotic Tactile Sensing (2019) — Tactile processing informed by human mechanoreception. Open source ↗

[24] A Flexible Triboelectric Tactile Sensor for Simultaneous Material and Texture Recognition (2021) — Macro/micro texture information from triboelectric tactile measurement. Open source ↗

[25] Multimodal Zero-Shot Learning for Tactile Texture Recognition (2024) — Cross-modal recognition of unseen texture categories. Open source ↗

## Vision / touch fusion

[26] Connecting Touch and Vision via Cross-Modal Prediction (2019) — Bidirectional learning from visual and tactile signals. Open source ↗

## Tactile representation learning

[27] Sparsh: Self-Supervised Touch Representations for Vision-Based Tactile Sensing (2024/2025) — Pretrained optical tactile representations across multiple datasets and sensors. Open source ↗

[28] AnyTouch: Learning Unified Static-Dynamic Representation across Multiple Visuo-Tactile Sensors (2025) — Towards sensor-agnostic tactile features across devices. Open source ↗

[29] AnyTouch 2 (2026) — Dynamic, force-aware cross-sensor tactile representations. Open source ↗

## Whole-body tactile skin

[30] Realizing Whole-Body Tactile Interactions with a Self-Organizing, Multi-Modal Artificial Skin on a Humanoid Robot (2015) — Modular multimodal robot skin and body-level contact behavior. Open source ↗

[31] Real-Time Control of a Humanoid Robot for Whole-Body Tactile Interaction (2025) — Multiple skin modalities integrated into whole-body robot control. Open source ↗

[32] RobotSweater: Scalable, Generalizable, and Customizable Machine-Knitted Tactile Skins for Robots (2023) — Low-cost scalable knitted pressure-sensitive skin. Open source ↗

## Whole-body tactile skin / review

[33] Humanoid Electronic-Skin Technology for the Era of Artificial Intelligence of Things (2025) — Review of e-skin, multimodal sensing and humanoid integration. Open source ↗

[34] Systematic Review of Human-Inspired Electronic Skin Systems (2025) — Review highlighting heterogeneous capabilities of many human-inspired skins. Open source ↗

## Replaceable tactile skin

[35] ReSkin: Versatile, Replaceable, Lasting Tactile Skins (2022) — Magnetic tactile sensing designed for replaceable sensor skins. Open source ↗

[36] AnySkin: Plug-and-Play Skin Sensing for Robotic Touch (2024/2025) — Practical replaceable tactile sensing and cross-instance robustness. Open source ↗

## Dexterous skin

[37] D'Manus: A Robot Hand with Tactile Sensing Across Fingers and Palm (2024/2025) — Distributed magnetic sensing on a dexterous hand; simulation component. Open source ↗

## Human demonstrations

[38] Feel the Force: Human Demonstration with Tactile Gloves (2025/2026) — Human tactile/force measurements for learning force-aware robot behavior. Open source ↗

## Tactile world models

[39] SPOTS: Multi-Modal World Model for Physical Robot Interactions (2026) — Simultaneously predicts visual and tactile robot-pushing outcomes. Open source ↗

[40] TacForeSight: Force-Guided Tactile World Model for Contact-Rich Manipulation (2026) — Force-conditioned tactile latent prediction for manipulation policies. Open source ↗

[41] TACO: TActile World Model as a Self-COrrector for Scalable Robot Policy Post-Training (2026) — Synthetic visuotactile corrections for policy improvement. Open source ↗

[42] FeelWorld: Visuo-Tactile World Model for Hierarchical Contact Prediction and Planning (2026) — Joint visual, contact, tactile latent and slip prediction. Open source ↗

[43] ContactWorld: What Matters in Vision-Tactile World Models for Contact-Rich Manipulation (2026) — Benchmark on structured representations and world-model planning. Open source ↗

## Review

[44] Tactile Sensing for Dexterous In-Hand Manipulation in Robotics - A Review (2011) — Early broad overview of tactile sensing in manipulation. Open source ↗

[45] Tactile Sensing in Dexterous Robot Hands - Review (2015) — Sensors, grasping, force/slip detection and robotic tactile control. Open source ↗

[46] Vision-Based Tactile Intelligence for Robotics: Sensing, Learning, and Embodied Manipulation (2026) — Recent survey of visuotactile learning and systems. Open source ↗

### Concept / review

[47] The Potential of Haptic Foundation Models (2026) — Perspective on general action-conditioned haptic representations. Open source ↗

### Simulation documentation

[48] NVIDIA Isaac Sim Physics Fundamentals (ongoing) — Material friction, restitution and deformable simulation primitives. Open source ↗

[49] NVIDIA Isaac Lab Visuo-Tactile Sensor Documentation (ongoing) — Simulation of tactile normal and shear fields for elastomer contact. Open source ↗

[50] NVIDIA Isaac Sim Hydroelastic Contact Documentation (ongoing) — Distributed contact patches and hydroelastic contact model. Open source ↗

## 18. Compact technical glossary and assumptions

SDF. Signed distance field: a representation of shape and proximity to a surface.

Taxel. A tactile sensing element, analogous to a pixel but measuring contact-related quantities.

Normal force / pressure. Force perpendicular to the surface; pressure expresses force per area.

Shear traction. Tangential force per area across a contact patch; essential for slip and friction.

Incipient slip. Partial local slipping before an object undergoes macroscopic sliding.

Compliance. Degree of deformation under applied load; depends on geometry and material.

Viscoelasticity. Time/rate-dependent mechanical response that mixes elastic and dissipative behavior.

Hydroelastic contact. Contact representation with spatially distributed contact surface/pressure.

FEM. Finite-element method for approximating continuum deformation and stress.

MPM. Material point method for simulating deformable bodies with particle/grid representations.

Inverse physics / system identification. Inferring material or model parameters from observed interactions.

Latent tactile state. Learned compressed representation of tactile measurements.

Sim-to-real. Testing whether a policy/model trained or calibrated in simulation works on physical hardware.

Provenance. Record of where measurements came from, including sensor IDs, calibration and specimen information.

Important assumptions: Equations and asset schemas in this report are conceptual, not an established universal tactile standard. The label “somatosensory world model” expresses the long-term research goal. The idea that a hybrid physics-based model improves transfer is a proposed experimental hypothesis. Individual published metrics are quoted only within their original experimental scopes. An absence of a fully integrated system in the reviewed works does not prove that no undiscovered project exists.

> **Final perspective. The opportunity is to connect three presently separated advances: physics-based contact, real measured tactile material properties, and learned future tactile prediction. A usable first milestone is a verified material/contact benchmark. Full-body synthetic touch should be treated as the ultimate systems challenge, pursued once transfer across actions and sensors is credible.**